

Project Documentation

Project Executable Files

Date: 10 July 2026

Team ID: SWTID-2026-6119

Project Name: Credit Card Approval Prediction System

Maximum Marks: 2 Marks

Purpose:

This document lists every executable and runnable file in the Credit Card Approval Prediction System repository, along with instructions to set up and run the project from scratch.

Executable Files

File	Purpose	How to Run
Flask/app.py	Main entry point of the web application. Loads the trained model and serves the prediction form and results.	python app.py
Notebooks/EDA_and_Model_Training.ipynb	Regenerates the trained model from raw data - EDA, preprocessing, training, evaluation, and saving model artifacts.	jupyter notebook (or Run All in Jupyter/VS Code)
requirements.txt	Lists all Python package dependencies required to run the project.	pip install -r requirements.txt

Setup & Run Instructions

- Step 1: Clone the repository and navigate into the project folder.
- Step 2: (Recommended) Create a virtual environment: `python -m venv venv`, then activate it.
- Step 3: Install dependencies: `pip install -r requirements.txt`
- Step 4: (Optional - model is already trained and saved) To regenerate the model, open and run all cells in `Notebooks/EDA_and_Model_Training.ipynb`.
- Step 5: Navigate into the Flask folder: `cd Flask`
- Step 6: Run the application: `python app.py`
- Step 7: Open a browser and go to `http://127.0.0.1:5000`

Live Application - Prediction Form:

Live Application - Prediction Result:

Example run with a strong applicant profile (high income, no prior default, employed):

➤ run with a strong applicant profile (high income, no prior default, employed):

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✓

Application Approved

Based on the details provided, the model predicts this application is likely to be approved.

Model confidence (probability of approval): 99.56%

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Screenshot captured from a live local run of the application, confirming the model returns a real Approved/Rejected prediction with a confidence score.